Blastsmart app: Pseudocode

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Function surface_blasting_calculator:
Input S/B ratio
Input explosive_density
Input blasthole_diameter
Input bench_height
Input stemming_factor

stemming_length= stemming_factor*(blast_hole_diameter/1000)
charge_length= bench_height – stemming_length
linear_charge_mass=(explosive_density* (blast_hole_diameter**2))/1273
mass_explsv_hole= linear_charge_mass* charge_length
Input technical_powder_factor
if technical_powder_factor is between 0.7 and 0.70:
    burden=  ((linear_charge_mass * (bench_height - stemming_length)) / (S/B
    ratio *    technical_powder_factor * bench_height)) ** (0.5)
    spacing=burden*( S/B ratio)
    subdrill=(1/3) * burden
    +ve/-ve_subdrill= input("Is the sub_drill +ve or -ve (+,-,0):")
    if +ve/-ve_subdrill is negative then:
        actual_powder_factor = =  ((linear_charge_mass * (bench_height  -
        stemming_length - subdrill)) / (burden *  spacing  * bench_height))
        return actual_powder_factor
    elif +ve/-ve_subdrill is positive then:
        actual_powder_factor = =  ((linear_charge_mass * (bench_height  -
        stemming_length + subdrill)) / (burden *  spacing  * bench_height))
        return actual_powder_factor
    else:
        actual_powder_factor = =  ((linear_charge_mass * (bench_height  -
        stemming_length)) / (burden *  spacing  * bench_height))
        return actual_powder_factor

    return stemming_length
    return burden

    Input relative_effective_energy
    Input rock_factor_constant

    If rock_factor_constant is between 6 and 14:
        mean_fragmentation_size= rock_factor_constant *
        ((technical_powder_factor) ** (-0.8)) * (mass_explsv_hole ** 0.167) *
        ((115 / relative_effective_energy) ** 0.633)
        return mean_fragmentation_size
```

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        else:
            return Invalid Rock Constant
else:
        return Invalid Technical powder factor
End function
```